

QA Cafe Fall 2026

Research briefing on the Fall 2026 newsletter: CDRouter Stability & Memory, Packet Viewer SmartFilter, Wi-Fi 7 testing, and customer stories

Prepared October 1, 2026 | Compiled from the newsletter email and from QA Cafe's own site, support documentation, and press releases

1. Executive summary

The Fall 2026 QA Cafe newsletter [1] has two product stories and two proof points. The first product story is the **CDRouter Stability & Memory Expansion**, which automates long-duration soak testing of broadband and Wi-Fi devices. The second is **SmartFilter** in Packet Viewer, which turns plain-English questions into validated Wireshark display filters through a customer-hosted AI webhook. The proof points are a CloudShark Enterprise customer story and a Wi-Fi 7 testing webinar.

Key findings

- **Soak testing is now packaged.** The expansion ships ready-to-run stability tests that repeat a stress action for anywhere from one minute to one week, take timestamped measurements, and judge the device against pass/fail rules [2][3].
- **Memory is read through standards, not agents.** Device memory is read from the TR-181 MemoryStatus object over USP (TR-369) or CWMP (TR-069), so no custom instrumentation is needed on the device under test [3].
- **SmartFilter never calls an AI provider itself.** Packet Viewer posts the user's question to an HTTP endpoint the customer runs, validates the returned filter against Wireshark, and retries once on failure. The feature is labeled experimental [5].
- **Wi-Fi 7 is a 2026 focus.** QA Cafe added an Advanced Wi-Fi 7 Module on July 13, 2026, and the newsletter promotes a webinar on practical Wi-Fi 7 validation [7][8].

What could not be verified

The full text of the InterConnect Defense case study and the Wi-Fi 7 webinar recording were not retrievable, so those two items are described only as far as the newsletter and QA Cafe's resource listings describe them (sections 5 and 6). Everything else below comes from QA Cafe's own pages.

2. About QA Cafe

QA Cafe describes itself as a software company of networking, consumer-electronics, and security specialists, founded in 2001, that builds network device test solutions and network analysis tools [14]. Its three main product lines are:

Product	What it does
CDRouter	Automated test solution for broadband gateways, Wi-Fi routers and mesh systems, and other CPE. Covers functional, conformance, performance, security, and management-protocol testing; also the official automated tool for Broadband Forum TR-069 and USP/TR-369 testing [3][11].
CloudShark Enterprise	Browser-based, centralized packet-capture analysis with Wireshark-compatible viewing plus Zeek and Suricata integration, unlimited seats and instances under a simple license [13].
Packet Viewer	Embeddable React components plus a containerized Wireshark-based backend, for vendors who want packet analysis inside their own cloud-managed product. It is the engine behind CloudShark [6].

The newsletter introduces the 2026 team, including new software developer Brett Dupont, who is building new test and packet analysis capabilities for all three product lines [1].

3. CDRouter Stability & Memory Expansion

3.1 The problem it targets

Many costly defects in gateways and Wi-Fi routers do not show up in functional or performance tests. Memory leaks, CPU exhaustion, and resource fragmentation build up slowly, and can end in poor throughput, failed features, crashes, or unexpected reboots after days or weeks in a customer's home [2]. QA Cafe's CEO, Erica Johnson, frames it as the gap between knowing how a device works at one moment and knowing how it behaves over time [2]. The newsletter adds that the expansion can surface in a day field issues that might take months to appear in the real world [1].

3.2 What the expansion does

- Repeats a functional stress action for a set duration while watching for degradation, rather than just generating sustained traffic [2][3].
- Cycles WAN, LAN, and PPPoE connections and checks that Wi-Fi clients stay associated under continuous traffic [3].
- Measures throughput against the device's own baseline, so no expected rate has to be configured in advance [3].
- Reads device memory from the TR-181 Device.DeviceInfo.MemoryStatus object over USP or CWMP while the device is under load [3].
- Correlates resource use with pass/fail results, performance measurements, test logs, and packet captures to speed up root-cause analysis [2].
- Supports IPv4 and IPv6, single-client and multi-client runs, and run lengths from one minute to one week [3].

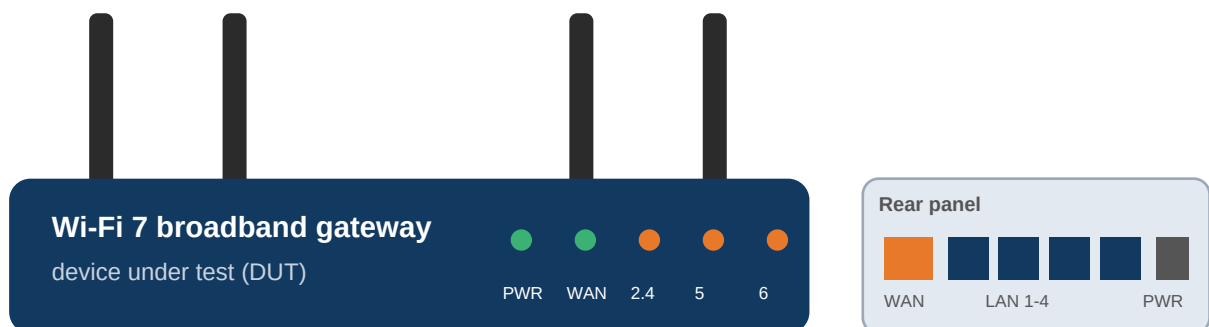


Figure 1. Illustration of a typical broadband gateway: front view with antennas and status LEDs (left), rear panel with typical ports (right). Schematic, not a photo of a specific product.

3.3 How a test run works

Most stability tests follow the same four-step pattern [3]:

- **Baseline.** CDRouter takes a first measurement before functional testing starts: a LAN throughput rate or the device's memory utilization.
- **Stress loop.** For the configured duration, CDRouter repeats the test's functional action and verifies each iteration succeeds.
- **Sampling.** After each interval of functional testing, it takes a timestamped measurement and checks it against the pass criteria. Timestamps show *when* a device degraded, not only the start and end values.

- **Final check.** At the end it takes a last measurement and compares it with the baseline.

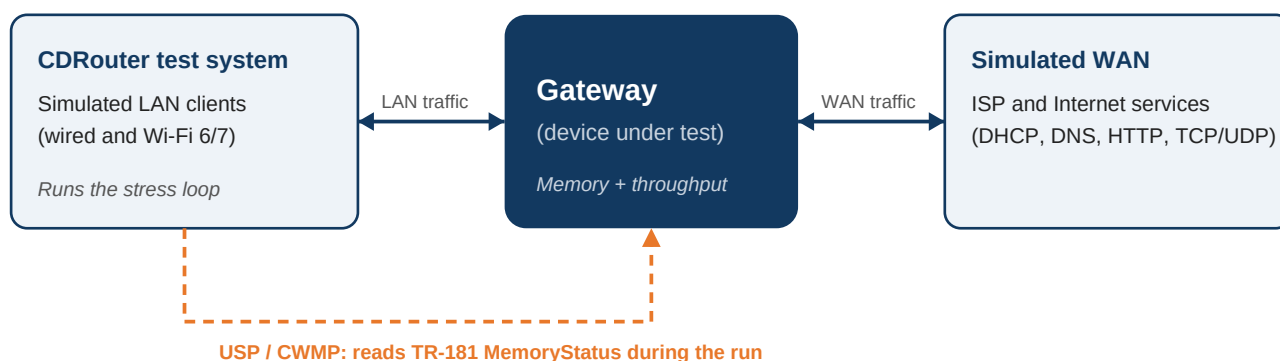


Figure 2. Where the gateway sits in a CDRouter stability test. No agent is installed on the gateway; memory is read through the standard management protocol.

3.4 Three test categories and their pass/fail rules

Category	What is judged	Default rule
Connectivity	Whether every link cycle (WAN, LAN, PPPoE) recovers, or the Wi-Fi client stays associated. No measurements are taken.	Any failed cycle counts against the run.
Throughput stability	Each sample against the test's own baseline. Measurement is always a UDP download using four streams, which keeps samples comparable.	Fails if a sample falls more than 10% below baseline.
Memory stability	Percentage of the device's physical memory in use, read over USP or CWMP. Judged as an absolute value, not against the baseline.	Fails if any one sample exceeds 80%.

A failed functional iteration counts against the run regardless of the measurements, and a run keeps going to its full duration by default so the results show whether a drop recovered [3]. A worked example in the guide: a 940 Mbps baseline gives a floor of 846 Mbps. A sample of 900 Mbps passes, 846 Mbps passes exactly at the floor, and 820 Mbps fails with a 12.8% drop [3].

3.5 Functional stress actions

Each stability test repeats exactly one action, named in the test case itself (for example, `stability_memory_usp_dhcp_restart`) [3]:

Action family	What is repeated
DHCP / IPv6	Restart or release the LAN client's DHCP client; re-acquire an IPv6 address by SLAAC or DHCPv6, verifying the address comes from the configured pool.
Wi-Fi	Disassociate and reassociate the wireless client, either to the same BSSID or to any BSSID.
TCP / HTTP / DNS	Open a large single TCP connection, transfer over HTTP, or send a large DNS query; fan-out variants hit several distinct WAN destinations.
Background traffic	Hold the device under one continuous TCP or UDP stream for the whole run (download or upload).

Multi-client versions of every family drive all configured LAN clients at once. Throughput is judged per client, so one starved client fails the test even if the others run at full speed. A memory reading taken while every client is busy can reveal growth that only appears under many clients [3].

3.6 Key settings and defaults

Setting	Default	Meaning
stabilityDuration	3600 s (max 604800)	How long each test repeats its stress action; one hour by default, up to one week.
stabilityFunctionalDuration	300 s	Stress time between measurements. Setting 0 measures only at start and finish.
stabilityPerfDuration	10 s	Length of each throughput measurement.
stabilityPerfMargin	10 %	Allowed throughput drop below baseline.
stabilityMemoryThreshold	80 %	Maximum memory in use before a sample fails.
stabilityFaultThreshold	0 (unlimited)	Faults tolerated before ending early; 1 gives a fail-fast smoke check.
stabilityTrafficType / Direction	TCP / download	Settings for the continuous background traffic only.

The guide's overnight memory example reads device memory over USP every 10 minutes for 12 hours and fails above 70% utilization [3]:

```
testvar supportsStabilityMemory      yes
testvar supportsPerformance          yes
testvar supportsUSP                  yes
testvar stabilityDuration             43200
testvar stabilityFunctionalDuration   600
testvar stabilityMemoryThreshold      70
```

3.7 Requirements and practical notes

- It is a **licensed expansion** that builds on the CDRouter Performance expansion and cannot be licensed without it [3].
- It needs an **NTA1000v5 or newer** test system from QA Cafe [3].
- USP memory modules need the USP expansion, CWMP modules need TR-069, IPv6 modules need IPv6, and multi-client modules need Multiport [3].
- The device must support the TR-181 MemoryStatus object, and a device that normally idles above 80% memory needs a higher threshold to avoid false failures [3].
- With default settings, one hour per test, a package of many stability tests can take days, so choose modules and durations deliberately [3].
- Each run produces a Stability and Memory dashboard in the CDRouter web interface with headline numbers (such as peak memory and lowest throughput), charts across the run, and a one-line verdict [3].
- Release timing: QA Cafe's press release is dated August 31, 2026, and states the expansion is available now [2].

4. Packet Viewer and SmartFilter

4.1 Packet Viewer in brief

Packet Viewer was launched on April 9, 2024 for cloud-managed network and security platforms that need to show customers packet traces without sending raw pcap files out of the vendor's cloud [6]. The front end is a set of React components compiled into the vendor's web app. The back end is a Docker-deployed service that connects to the vendor's own object storage (S3, NFS, or URLs) and uses Wireshark's protocol support and filter syntax [6]. The newsletter pitches the next step: bringing AI-assisted analysis into that embedded experience [1].

4.2 SmartFilter: how it works

SmartFilter lets a user type a question such as one about DNS errors or TCP resets from a given host, and translates it into a Wireshark display filter [5]. The design keeps AI choices on the customer's side:

- Packet Viewer itself ships with no model, provider, or API key. The customer provides an HTTP endpoint and enables it with the `--smartfilter-endpoint` flag, the `PV_SMARTFILTER_ENDPOINT` environment variable, or `config.yaml` [5].
- Credentials stay on the customer's infrastructure and never reach the browser, so the organization can change models whenever its AI strategy changes [5].
- In the front end, the `PacketViewer` view opts in with `enableSmartFilter`, which replaces the normal filter bar. Alternatively, a composable `SmartFilterInput` component can be placed in a custom layout [5].
- Without a configured endpoint, the capability is simply disabled [5].

4.3 Request flow

Step	What happens
1. Passthrough check	If the typed text is already a valid Wireshark display filter, it is applied directly and the webhook is never called.
2. Webhook call	Otherwise the server POSTs JSON containing the user's query and a release-versioned system prompt written by QA Cafe. Customers can forward that prompt to their model or ignore it.
3. Validation	The returned candidate filter is checked against Wireshark. A valid filter is applied to the packet list.
4. One retry	If invalid, the server calls the endpoint once more, including the rejected filter and Wireshark's reason, so the model can correct itself.

The endpoint answers with either a filter or a short user-facing explanation when it cannot produce one, and the explanation is shown as guidance rather than an error. Each call has a 30-second timeout. The interface reacts to HTTP status: 200 applies the filter or shows guidance, 503 means SmartFilter is not configured, 502 means the webhook failed (the only case where the user is invited to retry), and 500 signals an internal Packet Viewer problem [5].

The documentation marks SmartFilter as **experimental**, subject to change or removal, and asks customers for feedback [5].

5. CloudShark Enterprise: customer proof

The newsletter features a conversation with Seth Philips, Technical Lead at InterConnect Defense, LLC. According to the newsletter, CloudShark Enterprise changed how his organization works with packet captures, turning packet analysis from an expert-only activity into something hundreds of analysts now use daily [1]. The full case study was not retrievable for this briefing.

Related CloudShark evidence from QA Cafe's site, useful as context [13]:

- A global workflow-automation enterprise deployed CloudShark Enterprise across 24 data centers to keep packet data on-site for compliance, and used its graphical tools so junior analysts could find problem areas without losing expert-level detail.
- Trinity Cyber, a network security company, uses CloudShark Enterprise and its Threat Assessment tool in daily analysis work, and worked with QA Cafe to run it in Docker.
- A November 13, 2025 release added Large File Support for multi-gigabyte captures, integrated with Deep Search, Zeek, and Suricata workflows.

6. Wi-Fi 7 testing

The newsletter points to the recent webinar *Beyond Throughput: Practical Testing Strategies for Wi-Fi 7*, presented by QA Cafe Wi-Fi expert Brian Dubreuil (CWNA). It covers building an effective Wi-Fi 7 validation strategy and automating tests of the features that matter most [1][7]. Supporting material from QA Cafe's site:

- **NTA3000 platform (February 27, 2024):** the most advanced CDRouter platform, with tri-band Wi-Fi 6E / Wi-Fi 7 test capability and more 10G Ethernet ports [9].
- **Advanced Wi-Fi 7 Module (July 13, 2026):** adds more Wi-Fi 7 clients to CDRouter for realistic multi-client scenarios, including Multi-Link Operation (MLO), and lets teams combine Wi-Fi performance with long-term stability testing [8].
- **Testing scope:** QA Cafe's guidance stresses connectivity across 2.4, 5, and 6 GHz, security modes, MLO behavior, and performance with up to 115 clients across standards and bands, rather than radio-layer throughput alone [10].
- **Uniti Group case study:** Uniti reported replacing heavy, Wi-Fi-only systems with CDRouter, and cutting a device qualification cycle that previously took 6 to 8 weeks, with automated proof-of-concept cycles saving two weeks or more [11].

The link to section 3 is deliberate: QA Cafe positions Wi-Fi 7 testing and stability testing as one workflow, since issues such as memory growth under repeated Wi-Fi reassociation only appear under sustained load [3][8].

7. Epik Mellon podcast

Epik Mellon is QA Cafe's podcast, hosted by Jason Walls, Director of Technical Marketing. It focuses on the people and stories behind networking, standards, and cybersecurity, and is available on YouTube, Apple, Spotify, and Amazon [12][1]. The name is a play on a Greek phrase roughly meaning the future of communications [12]. Episodes listed in the newsletter:

Guest	Topic (as titled in the newsletter)
James Crawshaw, OMDIA / Informa Tech Target	"Up for a Little Light Reading"
Keith Parsons, Wireless LAN Professionals	"Wi-Fi Can Take You Anywhere"
Erika "The Dev" Dietrick	Reasonable thoughts about AI in network automation
Christopher LaPre, IntentIoT	"Putting the Smart in Smart Home"

Prospective guests can contact the host through the newsletter's invitation [1].

8. Timeline of related announcements

Date	Announcement
Feb 27, 2024	CDRouter NTA3000 platform with Wi-Fi 7 test capability [9]
Apr 9, 2024	Packet Viewer launched for cloud-managed network and security platforms [6]
Nov 13, 2025	CloudShark Enterprise Large File Support [13]
Jul 13, 2026	Advanced Wi-Fi 7 Module for CDRouter [8]
Aug 24, 2026	Stability and Memory User Guide published on the support site [3]
Aug 31, 2026	CDRouter Stability and Memory Expansion press release [2]
Fall 2026	Newsletter highlights SmartFilter, Wi-Fi 7 webinar, and CloudShark customer story [1]

9. Sources

- [1] QA Cafe, "The Fall 2026 QA Cafe Newsletter" (email supplied by the reader).
- [2] PR Newswire, "QA Cafe Launches CDRouter Stability and Memory Expansion to Fully Automate Soak Testing," Aug 31, 2026. prnewswire.com
- [3] QA Cafe Support, "CDRouter Stability and Memory User Guide." support.qacafe.com/cdrouter/user-guide/cdrouter-stability-memory-user-guide/
- [4] QA Cafe, CDRouter Expansions - Stability and Memory page. qacafe.com/test-solutions/cdrouter-benefits/expansions/
- [5] QA Cafe Support, "SmartFilter Webhook" (Packet Viewer deployment guide). support.qacafe.com/packet-viewer/deployment-guide/smartfilter-webhook/
- [6] QA Cafe, Packet Viewer pages and Apr 9, 2024 launch release. qacafe.com/analysis-tools/packet-viewer/ and qacafe.com/company/news/
- [7] QA Cafe Resources listing. qacafe.com/resources/
- [8] QA Cafe, "QA Cafe Adds Advanced Wi-Fi 7 Module for CDRouter," Jul 13, 2026. qacafe.com/company/news/
- [9] QA Cafe, "QA Cafe Releases New Test Solution for Advanced Broadband Gateways and Wi-Fi 7 Devices," Feb 27, 2024.
- [10] QA Cafe, "Qualifying Wi-Fi 7 devices using automated testing." qacafe.com/resources/qualifying-wi-fi-7-devices-using-automated-testing
- [11] QA Cafe case study, "How Uniti Group Accelerated Wi-Fi 7 Validation ... with CDRouter." qacafe.com/resources/
- [12] Epik Mellon podcast pages. qacafe.com/company/podcast and castbox.fm/channel/6884828
- [13] CloudShark Enterprise product and case-study pages; Nov 13, 2025 Large File Support release. qacafe.com
- [14] QA Cafe, "About QA Cafe." qacafe.com/company/about-qa-cafe/

Notes: [4] is cited for background only. Product details can change; check qacafe.com and support.qacafe.com for current information. This briefing paraphrases source material and is not an official QA Cafe document.